Multiresolution Analysis and the Wavelet Decomposition

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Key Concepts

1) The wavelet space $\mathcal{W}_j$ is the part of $\mathcal{V}_{j+1}$ that is not spanned by $\mathcal{V}_j$.

2) The wavelet space $\mathcal{W}_j$ is spanned by wavelets $\psi_{j,k}(t)$ over shifts k.

3) The wavelet $\psi_{j,k}(t)$ is defined in terms of the mother wavelet $\psi(t)$ as $\psi_{j,k}(t) = 2^{j/2}\psi(2^j t - k)$.

4) Wavelets are orthogonal to each other.

5) Since $\mathcal{W}_j \subset \mathcal{V}_{j+1}$, the mother wavelet must satisfy

$$\psi(t) = \sum_n h_1[n]\sqrt{2}\phi(2t-n)$$

6) The space at resolution J, $\mathcal{V}_J$, may be expressed in terms of a base resolution space $\mathcal{V}_0$ and the wavelet spaces.

$$\mathcal{V}_J = \mathcal{V}_0 \oplus \mathcal{W}_0 \oplus \mathcal{W}_1 \oplus \ldots \oplus \mathcal{W}_{J-1}$$

7) The wavelet decomposition for a signal in $\mathcal{V}_J$ is

$$x(t) = \sum_{k=-\infty}^{\infty} c_0[k]\phi_{0,k}(t) + \sum_{j=0}^{J-1}\sum_{k=-\infty}^{\infty} d_j[k]\psi_{j,k}(t)$$

a)

$$c_0[k] = \langle \phi_{0,k}(t), x(t) \rangle$$

b)

$$d_j[k] = \langle \psi_{j,k}(t), x(t) \rangle$$

8) The approximation waveform of $x(t)$ in $\mathcal{V}_{j_0}$ is

$$f_{j_0}(t) = \sum_{k=-\infty}^{\infty} c_{j_0}[k] \phi_{j_0,k}(t)$$

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9) The details waveform of $x(t)$ in $\mathcal{W}_j$ is

$$g_j(t) = \sum_{k=-\infty}^{\infty} d_j[k]\psi_{j,k}(t)$$

Video Screenshots

Multiresolution Analysis

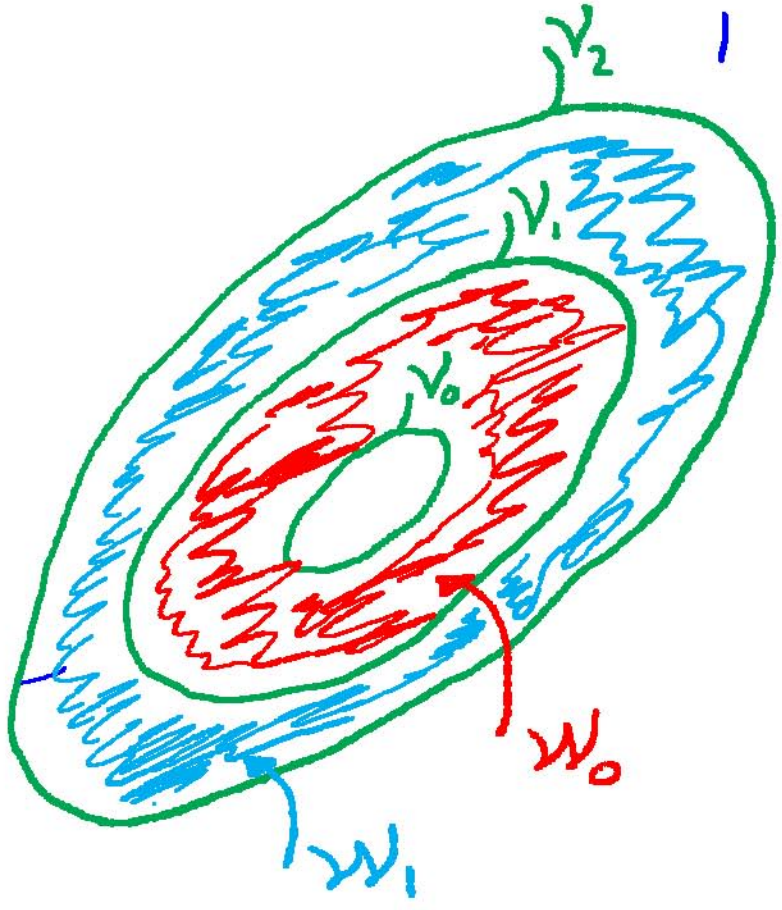
- scaling function $\phi(t)$

$$\phi_{j,k}(t) = 2^{j/2}\,\phi(2^j t - k)$$

V_j: $x(t) \in V_j \Rightarrow x(t) = \sum_k a_k \phi_{j,k}(t)$

Nesting $\ldots \subset V_{j-1} \subset V_j \subset V_{j+1} \subset \ldots$

$$\phi(t) = \sum_n h_0[n]\,\sqrt{2}\,\phi(2t-n)$$



Wavelet Space: W_j is part of V_{j+1} not in V_j

$\psi_{j,k}(t)$ spans W_j wavelets $\psi_{j,k}(t) = 2^{j/2}\psi(2^j t - k)$

Wavelets

- Orthogonality

$$\int \psi_{j,k}(t)\,\psi_{\ell,m}(t)\,dt = \begin{cases} 1 & j=\ell \text{ and } k=m \\ 0 & j\neq\ell \text{ or } k\neq m \end{cases}$$



$V_j \perp W_j$: $\int \psi_{j,k}(t)\,\phi_{j,\ell}(t)\,dt = 0$

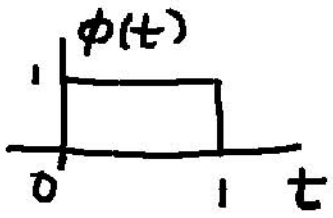
- $W_j \subset V_{j+1}$ $\quad \psi_{j,0}(t) = \sum_n h_1[n]\,\phi_{j+1,n}(t)$

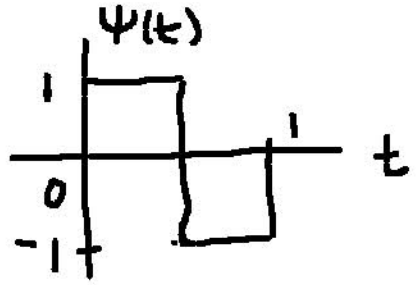
$$\Rightarrow \psi(t) = \sum_n h_1[n]\,\sqrt{2}\,\phi(2t-n)$$

Wavelet Spaces $W_j \perp W_\ell$ $j\neq\ell$

$$V_{j+1} = V_j \oplus W_j \Longrightarrow V_{j+1} = \overbrace{\underbrace{V_0 \oplus W_0}_{V_1} \oplus W_1}^{V_2} \oplus W_2 \oplus \cdots \oplus W_j$$

Example: Haar Wavelets

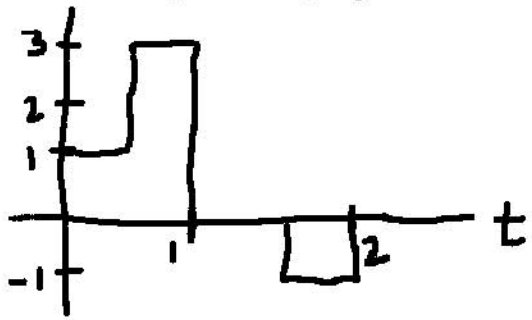




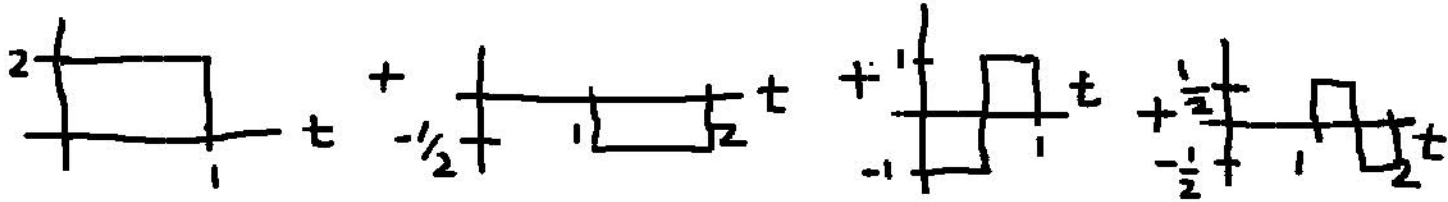
$$\psi(t) = \frac{1}{\sqrt{2}}\sqrt{2}\phi(2t) - \frac{1}{\sqrt{2}}\sqrt{2}\phi(2t-1)$$

$$h_1[0] = \frac{1}{\sqrt{2}}, \quad h_1[1] = -\frac{1}{\sqrt{2}}$$

$x(t) \in V_1$



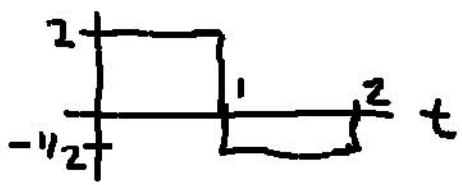
$=$



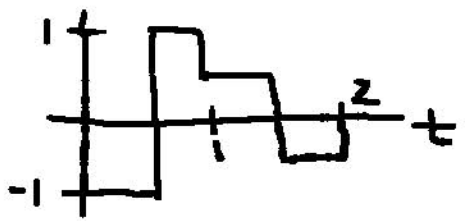
$$\phi(t) - \frac{1}{2}\phi(t-1) - \psi(t) + \frac{1}{2}\psi(t-1)$$

If $x(t) \in V_1 \Rightarrow x(t) = \underbrace{\sum_k a_k \phi(t-k)}_{V_0} + \underbrace{\sum_k b_k \psi(t-k)}_{W_0}$

$x(t) =$



$+$



Wavelet decomposition

$$V_J = V_0 \oplus W_0 \oplus W_1 \oplus W_2 \oplus \cdots \oplus W_{J-1}$$

$$x(t) \in V_J \Rightarrow x(t) = \underbrace{\sum_{k=-\infty}^{\infty} c_0[k]\, \phi_{0,k}(t)}_{\text{part in } V_0} + \sum_{j=0}^{J-1} \underbrace{\sum_{k=-\infty}^{\infty} d_j[k]\, \psi_{j,k}(t)}_{\text{part in } W_j}$$

Scaling coefficients: $c_0[k]$

$$c_0[k] = \langle \phi_{0,k}(t), x(t) \rangle = \int x(t)\, \phi_{0,k}(t)\, dt$$

Wavelet coefficients: $d_j[k]$

$$d_j[k] = \langle \psi_{j,k}(t), x(t) \rangle = \int \psi_{j,k}(t)\, x(t)\, dt$$

Discrete Wavelet Transform: $c_0[k]$, $d_j[k]$

Approximation and details

$x(t) \in V_J$

$$x(t) = \underbrace{\sum_k c_{j_0}[k]\,\phi_{j_0,k}(t)}_{f_{j_0}(t) \in V_{j_0}} + \sum_{j=j_0}^{J-1} \underbrace{\sum_{k=-\infty}^{\infty} d_j[k]\,\psi_{j,k}(t)}_{g_j(t) \in W_j}$$

$$x(t) = f_{j_0}(t) + \sum_{j=j_0}^{J-1} g_j(t)$$

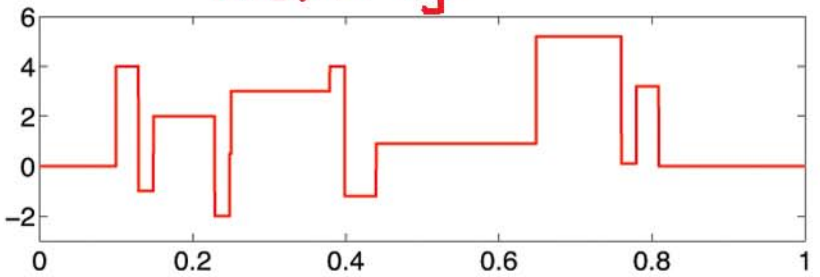
Approximation: $f_{j_0}(t) = \sum_{k=-\infty}^{\infty} c_{j_0}[k]\,\phi_{j_0,k}(t)$ (in V_{j_0})

Details: $g_j(t) = \sum_{k=-\infty}^{\infty} d_j[k]\,\psi_{j,k}(t)$ (in W_j)

Example:

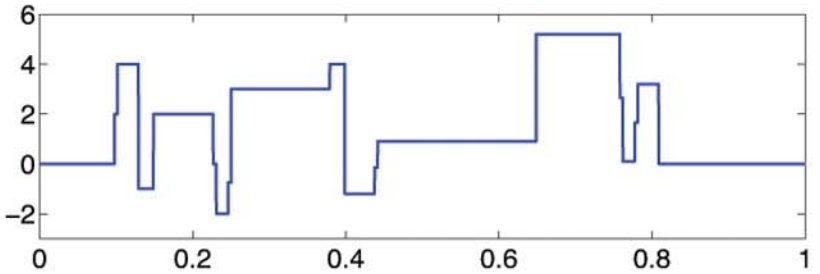
Signal

$x(t) \in V_J$

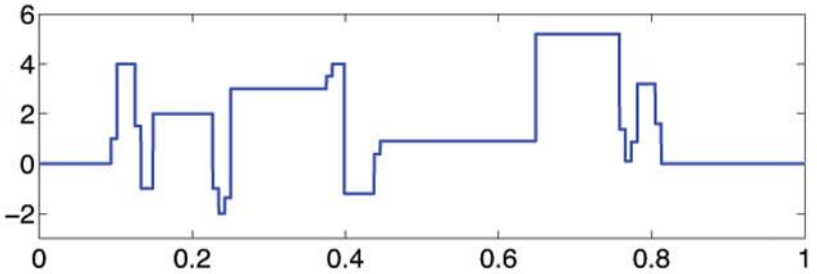


Approximation

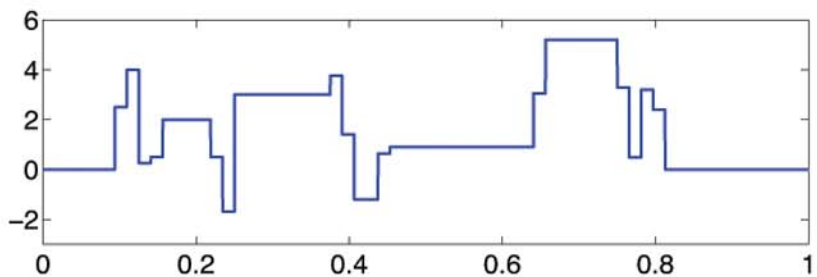
$f_{J-1}(t)$



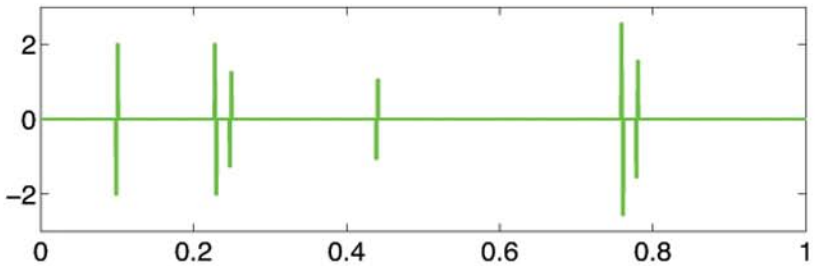
$f_{J-2}(t)$



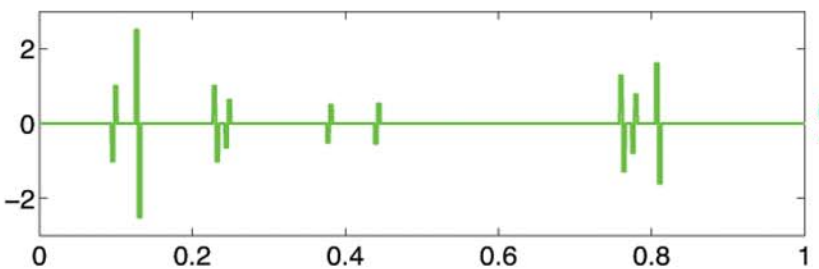
$f_{J-3}(t)$



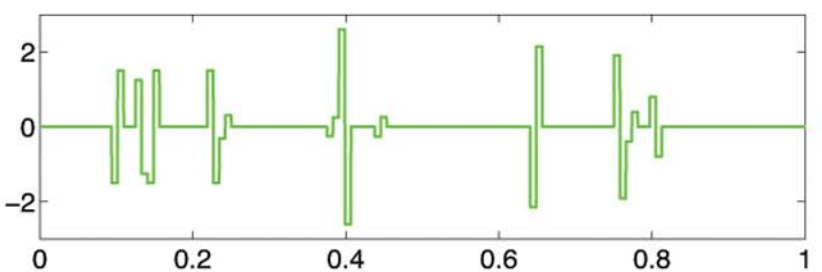
Details



$g_{J-1}(t)$



$g_{J-2}(t)$



$g_{J-3}(t)$

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